Rewarding Learning

General Certificate of Secondary Education
2026

Digital Technology

Unit 1:

(Compulsory Core)

Digital Technology

[GDG11]

MONDAY 18 MAY, AFTERNOON

MARK SCHEME

General Marking Instructions

Introduction

Mark schemes are published to assist teachers and students in their preparation for examinations. Through the mark schemes teachers and students will be able to see what examiners are looking for in response to questions and exactly where the marks have been awarded. The publishing of the mark schemes may help to show that examiners are not concerned about finding out what a student does not know but rather with rewarding students for what they do know.

The Purpose of Mark Schemes

Examination papers are set and revised by teams of examiners and revisers appointed by the Council. The teams of examiners and revisers include experienced teachers who are familiar with the level and standards expected of students in schools and colleges.

The job of the examiners is to set the questions and the mark schemes; and the job of the revisers is to review the questions and mark schemes commenting on a large range of issues about which they must be satisfied before the question papers and mark schemes are finalised.

The questions and the mark schemes are developed in association with each other so that the issues of differentiation and positive achievement can be addressed right from the start. Mark schemes, therefore, are regarded as part of an integral process which begins with the setting of questions and ends with the marking of the examination.

The main purpose of the mark scheme is to provide a uniform basis for the marking process so that all the markers are following exactly the same instructions and making the same judgements in so far as this is possible. Before marking begins a standardising meeting is held where all the markers are briefed using the mark scheme and samples of the students' work in the form of scripts. Consideration is also given at this stage to any comments on the operational papers received from teachers and their organisations. During this meeting, and up to and including the end of the marking, there is provision for amendments to be made to the mark scheme. What is published represents this final form of the mark scheme.

It is important to recognise that in some cases there may well be other correct responses which are equally acceptable to those published: the mark scheme can only cover those responses which emerged in the examination. There may also be instances where certain judgements may have to be left to the experience of the examiner, for example, where there is no absolute correct response – all teachers will be familiar with making such judgements.

Listing

Where **one** response is required to gain a mark, candidates will not gain credit if a correct response is given alongside one or more incorrect responses. This is referred to as listing.

				AVAILABLE MARKS
1	**(a)**	B	[1]	
	(b)	B	[1]	
	(c)	D	[1]	
	(d)	C	[1]	
	(e)	B	[1]	
	(f)	A	[1]	
	(g)	A	[1]	
	(h)	A	[1]	8
2	**(a)**	No marks are to be awarded for a description of Bit Rate. A higher bit rate improves sound quality or the reverse statement. [1] Any **one** from: As it captures the sound more accurately. [1] As it encodes more data (per second)/more bits per second. [1] **Note the question is about recording and not listening.**	[2]	
	(b)	Any **three** from: Changes sound waves into digital data. [1] Computer can only process/store/understand binary/digital data. [1] which is made of 1's and 0's. [1] Conversion allows the sound to be edited/processed by the computer. [1] Conversion allows sound to be played on digital devices/platforms. [1]	[3]	5
3	**(a)**	Portable Document Format	[1]	
	(b)	Rich Text Format	[1]	2

AVAILABLE MARKS

4 **(a)** MemberID [1]

(b)

Field Name	Data Type
ActivityID	Text/String [1]
DOB	Date/Time/Date/Short Date [1]
Price	Currency [1]

[3]

(c) (One to Many) [1]

(d) **(i)** A Query [1]

(ii) A Report [1]

(e) **(i)** Refers to large amounts of data sets [1]
Any **one** from:
Traditional database systems cannot handle effectively [1]
Have the potential to be mined (for information) [1]
All 3 together, Volume, Variety or Velocity [1]
NB do not award the mention of one or two V's, all three needed for a mark. [2]

(ii) The application of algorithms, programs or software [1]
Any **one** from:
Identify patterns, trends or relationships [1]
Helps organisations to process large volumes of data [1]
Analysing data to draw conclusions [1] [2] 11

5 **(a)** A **NETWORK** [1] is a group of computer systems linked together. They are connected usually by **CABLES** [1] or Wi-Fi. Computers networked together can share **RESOURCES** [1], such as files and printers, and are able to **COMMUNICATE** [1] with each other. [4]

(b) A Server/File Server [1]

(c) Any **two** from:
HTTP [1] HTTPS [1] FTP [1] [2]

(d)

Statement	Tick (✓)
Data travels in one direction.	✓ [1]
The main cable has terminators.	
Data collisions frequently occur in a ring network.	
Adding a workstation will disrupt the network.	✓ [1]
All workstations have equal access.	✓ [1]

[3] 10

			AVAILABLE MARKS
6	**(a)**	An **ABSOLUTE** [1] cell reference in a spreadsheet stays fixed on a **SPECIFIC** [1] cell, even when the **FORMULA** [1] is copied to other cells. [3]	
	(b)	Validation [1]	
	(c)	E4*F4/=(E4*F4)/Product(E4,F4) [1] NB must have comma. [1]	
	(d)	Replicate/Replication/Replicating [1]	
	(e)	=AVERAGE(G4:G8) [1] **or** =sum(G4:G8)/5 [1] [1]	
	(f)	Any **two** from: To see what impact a change in data entered has on other data in the spreadsheet [1] used to predict outcomes under different conditions [1] suitable example of 'what if' [1] [2]	
	(g)	1: C3/C4:C8 [1] Allow the word to 2: E3:E8 [1] 3: G3:G8 [1] [3]	12
7	**(a)**		

Facts	**Tick (✓)**
The CPU performs calculations	✓ [1]
The CPU stores large amounts of data permanently	
The CPU can run software without loading it into memory	
The CPU processes information provided by input devices	✓ [1]
The CPU runs the operating system and application software	✓ [1]

[3]

	(b)	The CPU/Computer can process more instructions [1] Any **one** from: Leads to better performance/can enhance CPU's performance [1] Faster execution of tasks [1] [2]	
	(c)	**(i)** ROM [1]	
		(ii) RAM [1]	7

8 Quality of written communication will be assessed in this question.

AVAILABLE MARKS

Level 0 ([0])

Answer is not worthy of credit.

Level 1 ([1]–[2])

The candidate refers to one [1] or two [2] of the advantages of e-Commerce in the correct context. The candidate makes limited use of spelling, punctuation, and grammar. The meaning of the text is not always clear. The candidate displays a limited form and style appropriate to the question. The organisation of the answer is limited.

Level 2 ([3]–[4])

The candidate describes one [3] or two [4] of the advantages of e-Commerce in the correct context. The candidate makes satisfactory use of spelling, punctuation, and grammar. The meaning of the text is usually clear. The candidate demonstrates a satisfactory form and style appropriate to the question. The organisation of the answer is satisfactory.

Level 3 ([5]–[6])

The candidate fully describes both advantages of e-Commerce and are used in the correct context. The candidate uses a good standard of spelling, punctuation, and grammar. The meaning of the text is always clear. The candidate demonstrates a good standard of form and style appropriate to the question. The organisation of the answer is good.

Possible Answers

1. Changes in Shopping Behaviour:

Convenience:

- E-commerce lets people shop anytime and anywhere, saving time and effort, especially for busy families.

Choices and Accessibility:

- Online shopping offers a wider selection of products and brands, including those not available nearby. Consumers can easily compare prices and find better deals.

2. Customer interactions:

Online Communities:

- E-commerce platforms offer social features that allow customers/users to share reviews and experiences, fostering a sense of community around brands and products.

Broader Social Connection:

- E-commerce enables people to connect online, expanding opportunities for engagement/interaction beyond geographic limits, and allowing more diverse engagement with communities and businesses worldwide.

NB. Candidates can describe **one** area from **Changes in Shopping Behaviour** and **one** area from **Customer interaction** and achieve the Level Marks.

Both points in each area do not need to be described. [6] 6

AVAILABLE MARKS

9 **(a)**

Responsibility	Tick (✓)
Provide suitable lighting	✓ [1]
Ignore trailing wires	
Provide ergonomically designed furniture	✓ [1]
Make employees pay for regular eye tests	
Keep equipment in good repair	✓ [1]

[3]

(b) Any **two** from:
Repetitive Strain Injury (RSI)/Carpal tunnel. [1]
Eye Strain [1]
Back/Neck Strain [1] [2]
NB. None of the following to be awarded. Sore eyes, Back issues, Sore back, Back injury, Back pain.

(c) Any **three** from:
An adjustable chair/desk/ergonomic chair/desk [1]
Ergonomic Keyboard/Mouse [1]
The monitor should be at eye level/non-glare/antiglare/adjustable brightness [1]
Organised cables (to avoid clutter and hazards) [1]
Wrist support/footrest [1]
Adequate lighting [1] [3] 8

10

Definition	Term
Manages communication between the file server and network computers	Switch [1]
The speed at which data is created, handled, and examined	Velocity [1]
A location in memory where data or information is stored temporarily	Register [1]
Unauthorised access to computer systems	Hacking [1]
Software that lets a user monitor another person's computer activity	Spyware [1]
Stores frequently accessed programs or instructions	Cache [1]
A file format where data is separated by comma	CSV [1]
Produces high quality printouts very quickly	Laser Printer [1]
A computer program that predicts how a real-life system might behave	Simulation [1]
Communicates using radio frequencies	Bluetooth [1]

[10] 10

11 (a)

Statement	Processing Method
Used for billing systems	Batch
Used in life critical systems, e.g. air traffic control	Real Time
Processes data instantly as it arrives	Real Time
Data is processed and completed later	Batch

Place the tick (✓) on the right of the Definition [4]

(b) Any **two** from:
Manages hardware [1]
Manages CPU/processor [1]
Manages peripherals [1]
Manages storage/secondary storage/hard disk [1]
Manages software [1]
Manages memory [1] [2]

(c) Utility programs help to maintain and **MANAGE** [1] the computer system. An example of a utility program is a **VIRUS** [1] scanner, which helps to protect against malware. A disk **DEFRAGMENTER** [1] is a utility program that optimises how data is stored on a hard drive. [3]

(d) Any **two** from:
Protects against data loss/(in case data gets lost) [1]
Allows lost/corrupt data to be restored [1]
Minimises downtime or loss [1] [2]

Total

AVAILABLE MARKS

11

90